

AISHWARYA MANDYA YOGANANDA

New York | +1 (607) 312-8260 | aishwaryamy20@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Gen AI enthusiast with 2+ years of experience in Software Engineering, Machine Learning, and NLP. Proficient in building **RAG pipelines**, fine-tuning LLMs, and hands-on experience with agentic frameworks and tools like **LangGraph**, **LangChain**, **Hugging Face**, and vector databases, delivering production-grade AI systems.

EDUCATION

Binghamton University, State University of New York

Aug 2024 – May 2026

Master of Science in Computer Science

TECHNICAL SKILLS

Programming: Python, C, C++, Java, JavaScript, SQL, Microservices

AI/ML Frameworks & Libraries: Pandas, NumPy, Matplotlib, Claude API, Claude Code, Anthropic SDK, GitHub Copilot, OpenAI API, LangChain, LangGraph, MCP, Langfuse, RAG, Prompt Engineering, Hugging Face, Whisper, NLP, Feature Engineering, Model Fine-tuning, FastAPI, Flask, Scikit-learn, TensorFlow, PyTorch, OpenCV, Keras

DevOps & Cloud: Docker, Kubernetes, Spark, PySpark, CI/CD, AWS S3, EC2, IAM, Lambda, GCP Compute Engine, VertexAI

Databases: PostgreSQL, MySQL, SQLite, MongoDB, Supabase, Google BigQuery

Project Management & Version Control: Jira, Asana, Agile, Git

PROFESSIONAL EXPERIENCE

Machine Learning Software Engineer | Manifoldz, NY, USA

May 2026 – Present

- Built a **hallucination-resistant, source-cited maritime AI assistant** using a **Python RAG** pipeline on a local **open-source LLM (Llama 3.2)**, grounding **1,700+ AIS vessel-tracking records** and trimming per-query API costs.
- Expedited onboarding time by **40%** by developing **Python** pipeline modules for the MSC barge survey tool, integrating **Whisper** transcription and **Gemini API** report generation via a custom **MCP server** with automated audio-dataset retrieval from the database.
- Strengthened issue traceability by identifying, categorizing, and logging **15+ software defects and data inconsistencies** across development workflows, cutting incident triage time for the engineering team.

Machine Learning Software Engineering Intern | Manifoldz, NY, USA

Sep 2025 – Apr 2026

- Trained **Whisper** in **Python** with **Hugging Face Transformers**, applying data augmentation and preprocessing optimizations to lower production ASR error rates by **43.3%**.
- Reduced routing delays by **30%** with a real-time **Python** pipeline that ingested **NWS weather data** and generated **OpenAI-powered route recommendations**.
- Upgraded model and alert performance by retraining **TensorFlow models** on curated domain data and applying feature engineering and validation, increasing accuracy by **18%** and reliability by **20%**.

Research Assistant | Binghamton University, Binghamton, NY

Jan 2025 – Apr 2025

- Engineered a secure real-time data pipeline on **FPGA (C, RTOS, AES-128 encryption)** for downstream inference workloads, achieving **99.9% reliability** in resource-limited environments.
- Regulated streaming latency for real-time inference workloads by profiling data pipelines, tuning system performance, and conducting load tests, sustaining **100 FPS throughput** and **95% uptime**.

Data Scientist Intern | BrainOVision Solutions, Bangalore, India

Apr 2022 – May 2022

- Streamlined data preprocessing, feature validation, and quality checks in **Python, Pandas, NumPy, and SQL**, cutting data preparation time by **30%** for downstream modeling.
- Developed an **NLTK sentiment pipeline** that automated text classification, feature extraction, and model scoring for real-time business insights, reducing manual analysis time by **50%**.

PROJECTS

AI Contact Enrichment Agent | **Next.js, TypeScript, GPT-4o, Supabase, Vercel**

[Demo](#) | 

- Condensed contact-research time to **under 45 seconds** with a conversational AI agent (**Next.js, TypeScript, GPT-4o**) that enriches B2B contacts with live intel and ICP scoring, persisting results in Supabase.
- Devised a **GPT-4o + Supabase pipeline** automating personalized outreach messages from live research, reducing per-prospect outreach time by **95%**.

Quantamind Legal AI Agent | **Python, Mistral 7B, LoRA, FAISS, RAG, Hugging Face**



- Designed an offline legal contract agent using **Mistral 7B, CUAD, FAISS, and RAG** to extract clauses and flag risks securely.
- Achieved **92.5% accuracy** with **LoRA fine-tuning** versus **85.3%** using few-shot prompting on the CUAD benchmark.

AI Incident Investigation Agent | **LangGraph, RAG, FastAPI, OpenAI, AWS**



- Constructed a semantic retrieval system using **FastAPI, PostgreSQL, and Redis**, compressing investigation time by **40%** and achieving **sub-200ms retrieval latency**.
- Facilitated **100+ daily queries** with **sub-2-second responses** by optimizing AWS serving, caching, and orchestration.
- Instrumented an automated **evaluation framework** with **Prometheus/Grafana** observability to monitor retrieval quality and system health in production.